Sao Aphisith Sithisack

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EDUCATION

University of Oklahoma **Norman, OK**
B.S. in Electrical Engineering (Honors) | GPA: 3.88 *Aug 2024 – May 2028*
Honors: Davis UWC Scholar (full tuition, 4 yrs) | Robert Vinton Clapp Scholarship
Relevant Coursework: Electronics, Microcontrollers, Circuits Lab, Digital Design, Digital Signal Processing, Electromagnetic Fields

EXPERIENCE

Sooner Competitive Robotics **Aug 2026 – Present**
Electrical Systems Engineer, Power Distribution *Norman, OK*

- Design the central **4-layer Altium PCB** for an autonomous IGVC 2027 robot: 12 V at up to 10 A, CAN, and E-Stop to 7 board ports, each with INA220 current sensing.
- Swapped the team's usual reverse-polarity Schottky for a P-MOSFET, cutting calculated loss at 8 A from **3.6 W to 0.7 W**; also specified the 12 V input's 10 A fuse and TVS diode.
- Replaced last year's 12 V to 3.3 V LDO with an AP63301 buck to reduce heat, for an estimated 55–70% efficiency at 50–100 mA (datasheet curve) versus 28% at most.
- Selected the **STM32C092** microcontroller and designed its circuit: CAN transceiver, I^2C to the 7 current sensors, SWD programming, USB-C to UART debug, and test points for bring-up.

Pulsed Laser Template Engineering (PLATEN) Lab **Jan 2026 – Present**
Undergraduate Researcher, Semiconductor Nanofabrication *Norman, OK*

- Fabricate high-aspect-ratio micro- and nanostructures in a cleanroom using photolithography, reactive ion etching (RIE), and pulsed laser deposition (PLD).
- Deposit functional oxide films by PLD and run XRD on multiple oxides, for research into IC fabrication without costly EUV lithography.

Student Learning Center **Aug 2025 – Present**
Peer Learning Assistant | 20 hrs/week *Norman, OK*

- Tutor 250+ students per semester in physics, calculus, and chemistry.

PROJECTS

TI MSPM0 USB-C Microcontroller Board **Aug 2026 – Sep 2026**
KiCad | 2-layer PCB | write-up *Personal Project*

- Designed the schematic for a USB-C powered **TI MSPM0G3507** board with a 3.3 V LDO, USB-to-UART bridge, ESD protection, 8 MHz crystal, and SWD debug.

Mixed-Signal Function Generator **Oct 2026 – Present**
KiCad | LTspice *Personal Project*

- Design a function generator targeting 1 Hz to 1 MHz in six ranges, with an op-amp integrator and Schmitt trigger oscillator plus an MCU-driven R-2R source, on a 4-layer KiCad PCB.
- Chose a discrete differential-pair sine shaper over a DDS chip and specified the output chain: waveform select, amplitude, DC offset, -20 dB attenuator, and 50 Ω buffer.
- Simulated the oscillator and sine shaper in LTspice; 20 **Monte Carlo** runs with 1% resistors gave a median THD of 1.08%.
- Planned MCU frequency counting, ADC amplitude readback, and an I^2C OLED display.

LEADERSHIP

United World College Lao PDR National Committee **Jul 2024 – Present**
Co-President & Alumni Coordinator *Remote / Laos*

- Co-lead a 20+ volunteer committee that runs Laos' UWC scholarship selection, narrowing 150+ applicants per cycle to 3–4 placements; led fundraising of $2,500+.

IEEE Student Branch at OU **Sep 2026 – Present**
Event Committee Member *Norman, OK*

- Plan event logistics and design circuits for technical events, including a 555-timer sound circuit and 12-board DigiKey BOM.

Association of Southeast Asian Nations at OU **Nov 2024 – May 2026**
Sponsorship & Volunteering Chair *Norman, OK*

- Raised $1,000+ through outreach to 20+ donors; organized cultural events.

TECHNICAL SKILLS

PCB & Hardware: Altium Designer (4-layer PCB), KiCad (schematic capture), power protection and distribution, USB-C, UART, CAN, I^2C, SWD, LTspice (Monte Carlo)
Fabrication & Lab: Photolithography, RIE, pulsed laser deposition, X-ray diffraction (XRD), cleanroom processing, oscilloscope, DMM
Programming: C, C++, Python, MATLAB, LabVIEW, Linux, Git